



Islamic University of Technology

Department of Computer Science and Engineering (CSE)

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Lecture 3

Simplifying Boolean Expressions: Karnaugh Maps

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1 K-Maps

The rules for simplifying Boolean expressions using Karnaugh maps (K-maps) with 2 to 6 variables and don't-care conditions.

1.1 Two-Variable K-Map

Map layout (variable A row, B column):

$A \backslash B$	0	1
0	m_0	m_1
1	m_2	m_3

Adjacency: Cells are adjacent horizontally, vertically, and with wrap-around (top/bottom, left/right). m_0 adjacent to m_1, m_2 ; m_3 adjacent to m_1, m_2 .

Group sizes(2^n): 1, 2, 4. A group of two must be adjacent cells. All four cells form a group of 4 (tautology).

Product term: For a group, eliminate variables that change; keep constant variables (0→complemented, 1→uncomplemented).

Example: grouping the right column ($B = 1$):

$A \backslash B$	0	1
0	m_0	m_1
1	m_2	m_3

A changes, $B = 1$ constant. Term: B .

1.2 Three-Variable K-Map

Layout (row A , columns BC in Gray code order 00,01,11,10):

$A \backslash BC$	00	01	11	10
0	m_0	m_1	m_3	m_2
1	m_4	m_5	m_7	m_6

Adjacency: Top/bottom rows are adjacent (wrap vertically). Left/right columns wrap (column 10 adjacent to 00).

Group sizes(2^n): 1,2,4,8. Valid shapes: horizontal 1×2 , vertical 2×1 , 2×2 square, a full row (1×4), or two adjacent rows ($2 \times 4 =$ whole map).

Product term: Eliminate variables that change inside the group.

Example: a 2×2 group covering $BC = 11, 10$ (right two columns):

$A \backslash BC$	00	01	11	10
0	m_0	m_1	m_3	m_2
1	m_4	m_5	m_7	m_6

Here A and B change, $C = 1$ constant.
Term: C .

Example: a 1×4 group covering $A = 0$ (top row):

$A \backslash BC$	00	01	11	10
0	m_0	m_1	m_3	m_2
1	m_4	m_5	m_7	m_6

Here B and C change, $A = 0$ constant.
Term: $\bar{A}$.

1.3 Four-Variable K-Map

Layout (rows AB , columns CD both in Gray code):

$AB \backslash CD$	00	01	11	10
00	m_0	m_1	m_3	m_2
01	m_4	m_5	m_7	m_6
11	m_{12}	m_{13}	m_{15}	m_{14}
10	m_8	m_9	m_{11}	m_{10}

Adjacency: Toroidal — top/bottom rows adjacent, left/right columns adjacent.

Group sizes(2^n): 1,2,4,8,16. The group must be rectangular with dimensions powers of two ($1 \times 1, 1 \times 2, 2 \times 1, 2 \times 2, 1 \times 4, 4 \times 1, 2 \times 4, 4 \times 2, 4 \times 4$).

Product term: Variables constant across the entire rectangle are kept.

Example 1: 2×2 group at $AB = 01, 11$
and $CD = 01, 11$

Example 2: 2×2 group (columns
 $CD = 11, 10$)

$AB \backslash CD$	00	01	11	10
00	m_0	m_1	m_3	m_2
01	m_4	m_5	m_7	m_6
11	m_{12}	m_{13}	m_{15}	m_{14}
10	m_8	m_9	m_{11}	m_{10}

$AB \backslash CD$	00	01	11	10
00	m_0	m_1	m_3	m_2
01	m_4	m_5	m_7	m_6
11	m_{12}	m_{13}	m_{15}	m_{14}
10	m_8	m_9	m_{11}	m_{10}

A and C change; $B = 1, D = 1$ constant.
Term: BD

A, B change, $C = 1$ and D changes. Term:
 C

Example 3: 1×4 group (row $AB = 00$)

Example 4: Wrap-around 2×2 group
(four corners)

$AB \backslash CD$	00	01	11	10
00	m_0	m_1	m_3	m_2
01	m_4	m_5	m_7	m_6
11	m_{12}	m_{13}	m_{15}	m_{14}
10	m_8	m_9	m_{11}	m_{10}

$AB \backslash CD$	00	01	11	10
00	m_0	m_1	m_3	m_2
01	m_4	m_5	m_7	m_6
11	m_{12}	m_{13}	m_{15}	m_{14}
10	m_8	m_9	m_{11}	m_{10}

B, C, D change, $A = 0$ constant. Term: $\bar{A}$

A and C change; $B = 0, D = 0$ constant.
Term: $\bar{B}\bar{D}$

1.4 Five-Variable K-Map

Use two 4-variable maps, one for $A = 0$ and one for $A = 1$, with rows BC and columns DE in Gray code.

$A = 0$					$A = 1$				
$BC \backslash DE$	00	01	11	10	$BC \backslash DE$	00	01	11	10
00	m_0	m_1	m_3	m_2	00	m_{16}	m_{17}	m_{19}	m_{18}
01	m_4	m_5	m_7	m_6	01	m_{20}	m_{21}	m_{23}	m_{22}
11	m_{12}	m_{13}	m_{15}	m_{14}	11	m_{28}	m_{29}	m_{31}	m_{30}
10	m_8	m_9	m_{11}	m_{10}	10	m_{24}	m_{25}	m_{27}	m_{26}

Adjacency: Each sub-map is toroidal. Additionally, cells with identical BC, DE coordinates in the two maps are adjacent (they differ only in A).

Group sizes(2^n): 1,2,4,8,16,32. Groups can be formed:

- Entirely inside one sub-map (same rules as 4-variable, max size 16).
- Spanning both sub-maps: take exactly the same rectangular group in both maps. Such a group doubles in size (e.g., a 2×2 in each map yields a group of 8) and eliminates A from the term.

Product term: For a group inside one map, treat as 4-variable case with A fixed (complemented if $A = 0$, uncomplemented if $A = 1$). For a group spanning both maps, drop A .

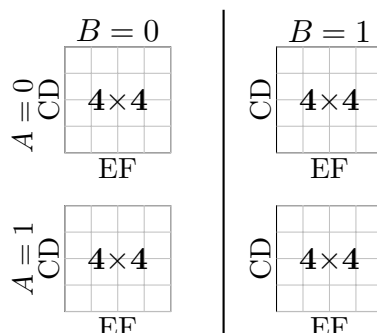
Example: a group spanning both maps formed by the 2×2 rectangle at $BC = 01, 11$ and $DE = 01, 11$:

$A = 0$					$A = 1$				
$BC \backslash DE$	00	01	11	10	$BC \backslash DE$	00	01	11	10
00	m_0	m_1	m_3	m_2	00	m_{16}	m_{17}	m_{19}	m_{18}
01	m_4	m_5	m_7	m_6	01	m_{20}	m_{21}	m_{23}	m_{22}
11	m_{12}	m_{13}	m_{15}	m_{14}	11	m_{28}	m_{29}	m_{31}	m_{30}
10	m_8	m_9	m_{11}	m_{10}	10	m_{24}	m_{25}	m_{27}	m_{26}

Since A varies and $B = 1, D = 1$ constant, term: BD .

1.5 Six-Variable K-Map

The map consists of four 4×4 sub-maps for the four combinations of two splitting variables (e.g., A and B) in Gray order: $(A, B) = (0, 0), (0, 1), (1, 1), (1, 0)$. Each sub-map uses rows CD and columns EF in Gray code. The sub-maps are arranged in a 2×2 grid:



Adjacency: Each sub-map is toroidal internally. Sub-maps that are horizontally or vertically adjacent in this grid correspond to changing only B or only A , respectively. Cells with identical CD, EF coordinates in two adjacent sub-maps are adjacent. Diagonally positioned sub-maps (changing both A and B) are not directly adjacent.

Group sizes: 1,2,4,8,16,32,64. Valid groups:

- Inside a single sub-map (max 16).
- Same rectangle in two horizontally adjacent sub-maps (same A , adjacent B) $\rightarrow$ eliminates B .
- Same rectangle in two vertically adjacent sub-maps (adjacent A , same B) $\rightarrow$ eliminates A .
- Same rectangle in all four sub-maps $\rightarrow$ eliminates both A and B . (The four sub-maps form a 2×2 block; a group spanning them is valid if the same rectangle is present in each.)

The overall group must be a subcube: identical rectangle in a rectangular set of sub-maps that are adjacent in the AB Gray-code plane.

Product term: Retain variables constant in the rectangle, omit those split variables that change across the combined sub-maps.

1.6 Don't-Care Conditions

Cells marked with X (don't-care) can be assigned either 0 or 1 to simplify the expression.

- Use an X as a 1 if it helps form a larger group (covers more cells, eliminates more variables).
- Do not create a group that consists solely of X 's (it must include at least one 1).
- Any X not included in a group is treated as 0.

During covering, don't-cares do not have to be covered; only the 1's must be covered.

1.7 Simplification Procedure

1. Draw the appropriate K-map and fill cells with 1, 0, or X from the truth table.
2. Encircle all **prime implicants** — largest valid groups that cover 1's (and possibly X 's) and cannot be enlarged further.
3. Identify **essential prime implicants**: those that cover a 1 not covered by any other prime implicant. These must appear in the final expression.
4. From the remaining prime implicants, choose a minimal set that covers all still-uncovered 1's. (Don't-cares may be left uncovered.)
5. Write the Sum-of-Products expression as the OR of the product terms corresponding to each selected group.

1.8 Examples

You can practice the following examples to solidify your understanding of K-map simplification, but your practice should not be limited to only these examples. Try creating your own truth tables and simplifying them using K-maps.:

- i) $F(A, B, C, D) = \sum m(0, 1, 2, 5, 7, 8, 9, 10, 13, 15)$
- ii) $F(A, B, C, D) = \sum m(0, 1, 2, 5, 6, 7, 8, 10, 13, 14, 15)$
- iii) $F(A, B, C, D) = \sum m(0, 1, 3, 5, 7, 8, 9, 11, 13, 15)$
- iv) $F(A, B, C, D) = \sum m(0, 1, 2, 5, 7, 8, 9, 10, 13, 15)$
- v) $F(A, B, C, D) = \sum m(0, 1, 2, 3, 5, 7, 8, 9, 11, 13, 15)$
- vi) $F(A, B, C, D) = \sum m(0, 2, 5, 7, 8, 10, 13, 15)$
- vii) $F(A, B, C, D) = \sum m(1, 3, 4, 6, 8, 9, 11, 13, 15), d(0, 2, 12)$
- viii) $F(A, B, C, D) = \sum m(1, 3, 4, 6, 8, 9, 11, 13, 15), d(0, 2, 14)$
- ix) $F(A, B, C, D, E) = \sum m(1, 3, 5, 7, 9, 11, 13, 15, 17, 19, 21, 23, 25, 27, 29, 31), d(0, 2, 4, 6)$
- x) $F(A, B, C, D, E) = \sum m(0, 1, 2, 3, 8, 9, 10, 11, 16, 17, 18, 19, 24, 25, 26, 27)$
- xi) $F(A, B, C, D, E) = \sum m(1, 4, 5, 8, 9, 12, 13, 16, 17, 20, 21, 24, 25), d(2, 3, 14, 15)$
- xii) $F(A, B, C, D, E) = \sum m(0, 2, 5, 7, 8, 10, 13, 15, 16, 18, 21, 23, 24, 26, 29, 31), d(1, 3, 14, 30)$
- xiii) $F(A, B, C, D, E) = \sum m(3, 7, 11, 15, 19, 23, 27, 31)$
- xiv) $F(A, B, C, D, E, F) = \sum m(1, 3, 7, 9, 15, 17, 23, 25, 31, 33, 39, 41, 47, 49, 55, 57, 63), d(0, 2, 4, 6)$
- xv) $F(A, B, C, D, E, F) = \sum m(8, 9, 10, 11, 24, 25, 26, 27, 40, 41, 42, 43, 56, 57, 58, 59)$
- xvi) $F(A, B, C, D, E, F) = \sum m(5, 6, 7, 12, 13, 14, 21, 22, 23, 28, 29, 30, 37, 38, 39, 44, 45, 46), d(1, 2, 3,$
- xvii) $F(A, B, C, D, E, F) = \sum m(0, 1, 2, 3, 16, 17, 18, 19, 32, 33, 34, 35, 48, 49, 50, 51), d(4, 5, 6, 7)$
- xviii) $F(A, B, C, D, E, F) = \sum m(10, 14, 18, 22, 26, 30, 34, 38, 42, 46, 50, 54, 58, 62)$
- xix) $F(A, B, C, D) = \prod M(0, 2, 5, 7, 8, 10, 13, 15), d(1, 3)$
- xx) $F(A, B, C, D) = \prod M(1, 3, 4, 6, 9, 11, 12, 14)$
- xxi) $F(A, B, C, D) = \prod M(0, 1, 4, 5, 8, 9, 12, 13), d(2, 3, 14, 15)$
- xxii) $F(A, B, C, D) = \prod M(2, 3, 6, 7, 10, 11, 14, 15), d(0, 8)$
- xxiii) $F(A, B, C, D) = \prod M(0, 3, 5, 6, 9, 10, 12, 15), d(1, 2, 4, 7)$

To be Noted

If given in Product of Maxterms and you are more comfortable in minterms, you should convert in the following way:

- Convert the maxterms ($\prod M$) into missing indices of the full set $\{0, 1, \dots, 15\}$.
- The missing indices represent minterms ($\sum m$).
- Don't cares (d) are ignored during conversion but can be used in simplification.

Example 1 (without don't care):

$$F(A, B, C, D) = \prod M(1, 3, 5, 7, 9, 11, 13, 15)$$

Total set: $\{0, \dots, 15\}$

Missing minterms:

$$\Rightarrow F = \sum m(0, 2, 4, 6, 8, 10, 12, 14)$$

Example 2 (with don't care):

$$F(A, B, C, D) = \prod M(0, 2, 5, 7, 8, 10), \quad d(1, 3)$$

Total set: $\{0, \dots, 15\}$

Step 1: Missing indices

$$\Rightarrow \{1, 3, 4, 6, 9, 11, 12, 13, 14, 15\}$$

Step 2: Separate don't cares

$$\sum m(4, 6, 9, 11, 12, 13, 14, 15), \quad d(1, 3)$$

If nothing is mentioned then you can give your solution in either Sum of Minterms or Product of Maxterms, but you should be consistent in your solution. You should not mix minterms and maxterms in the same solution.

But if one form is explicitly mentioned you have to give the answer in the format that has been asked. For Maxterms just group 0's instead of 1's and prioritize 0's (0 means original form and 1 means complemented form) instead of 1's (1 means original form and 0 means complemented form).

To be noted, there might be multiple correct answers for the same problem. You should not worry about that. The solutions must have same number of terms and literals. Just make sure your answer is correct and simplified as much as possible.

1.9 Tools:

[Karnaugh Map Solver](#)

For learning and practice, you should not use this during exams.